High-Dimensional Data: Challenges & Strategies

Workshop 5 — Module 1

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Data Visualization for Data Scientists
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Module Roadmap

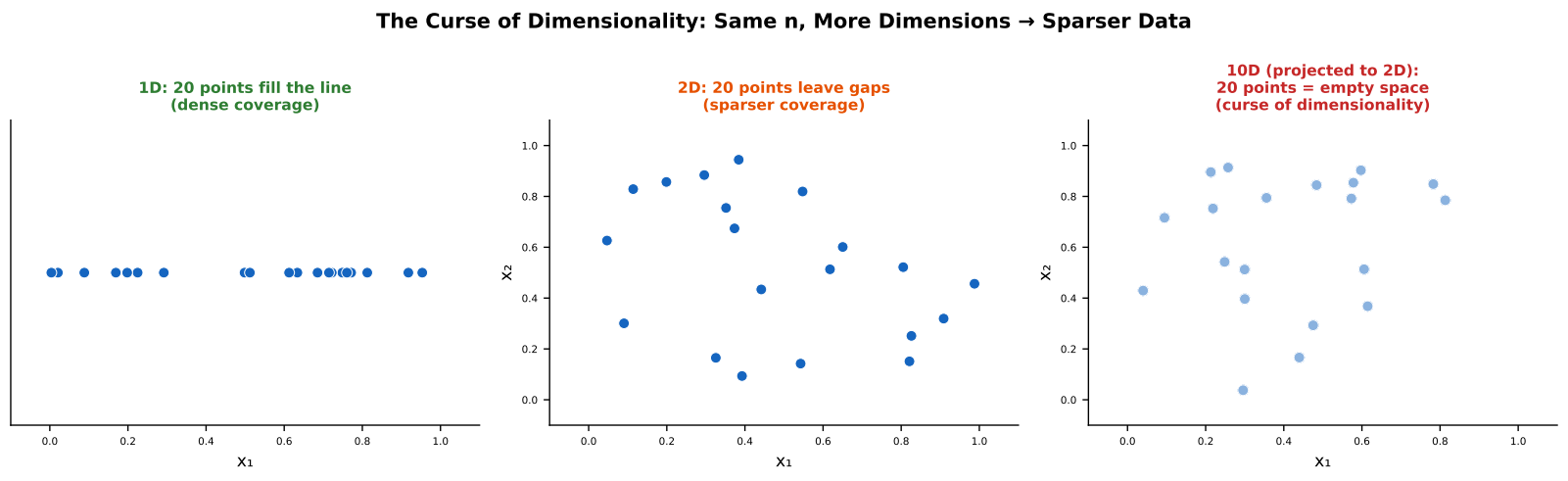
1. The curse of dimensionality: why high-D data is hard to plot
2. Visual encoding channels ranked by perceptual accuracy
3. Multi-encoded scatter: 5 variables on one chart
4. Aesthetic overload: when encoding fails
5. Five strategies for high-dimensional visualization
6. Aggregation techniques for large n

Learning Outcome

You will understand why standard 2D plots fail for high-dimensional data, apply multiple encoding channels without overload, and choose the right strategy (reduce, encode, facet, parallel, cluster) for any multivariate analysis.

The Curse of Dimensionality

Same n, More Dimensions $\rightarrow$ Sparser Data



Pro-Tip

With p dimensions and n points, the proportion of the unit hypercube covered falls as $n^{-1/p}$. With 20 points in 10D, each point is essentially alone — there are no "neighbours" to compare it with. This is why we need **dimension reduction** or **clever encoding** to visualize high-D data.

What Changes at High Dimensions?

Dimension	Standard Approach	High-D Challenge
1 variable	Histogram, boxplot	Still works (marginally)
2 variables	Scatter plot	Still works
3 variables	Scatter + colour/size	Starts getting cluttered
5–10 variables	Pairs plot (p^2 panels)	Becomes unwieldy
10–100 variables	???	Need dimension reduction
100+ variables	???	Need algorithmic reduction + summary vis

Visual Encoding Channels

Ranked by Perceptual Accuracy

Visual Encoding Channels: Ranked by Perceptual Accuracy

#	Channel	Accuracy	Chart Examples	Encodes
1	**Position (x, y)**	Most accurate (Cleveland & McGill)	Scatter, line, bar	2 vars
2	**Length**	Second most accurate	Bar height, line length	1 var
3	**Angle / Slope**	Moderate accuracy	Line slope, pie angle	1 var
4	**Area**	Weak: 2× area ≠ 2× perceived	Bubble size, treemap	1 var
5	**Colour hue**	Categorical only (≤8 distinguishable)	Group encoding	1 cat
6	**Colour saturation / luminance**	Quantitative, limited precision	Heatmap, choropleth	1 var
7	**Shape / Texture**	Categorical only (≤6 distinguishable)	Point shape, fill pattern	1 cat

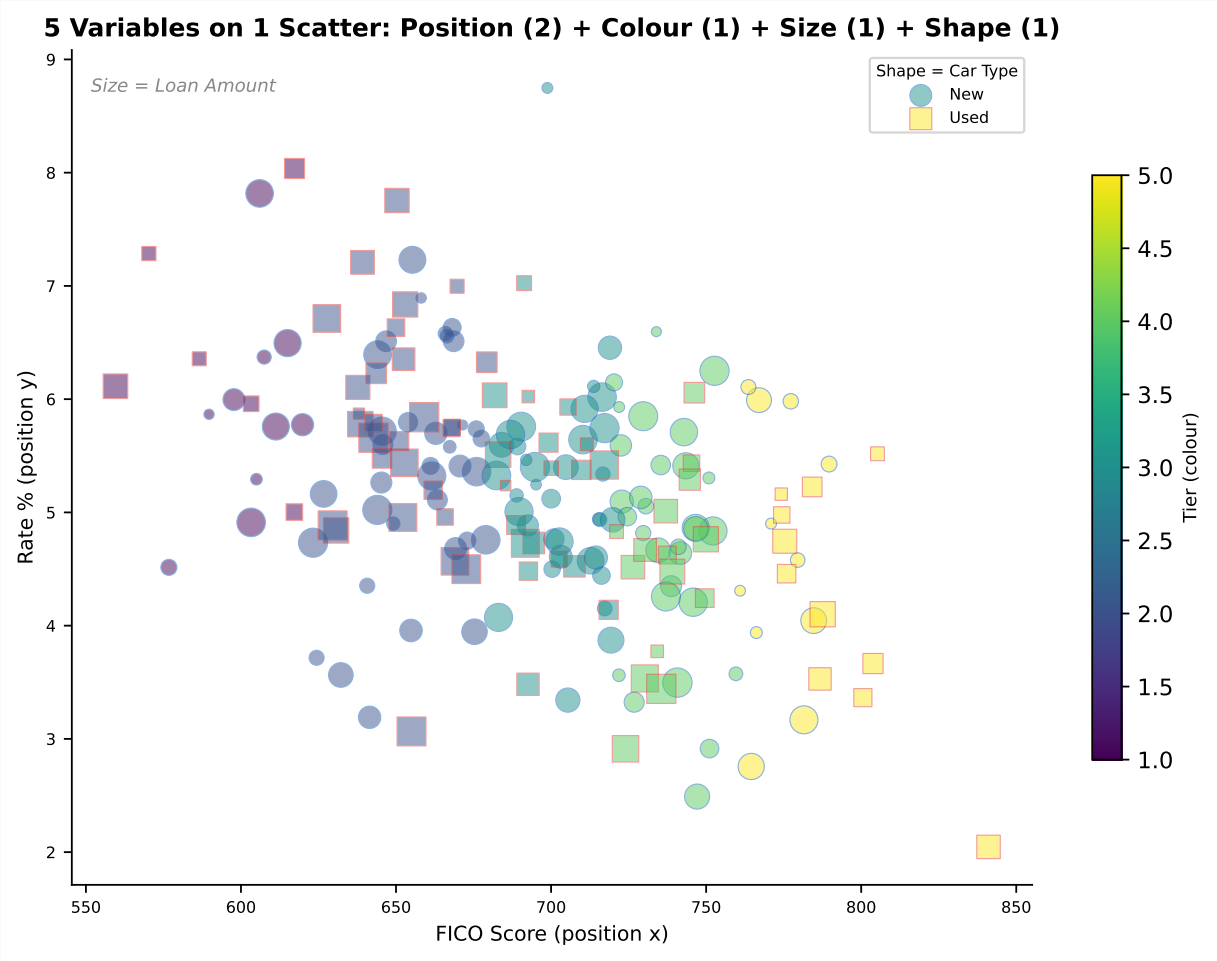
Encoding Budget: How Many Variables Per Chart?

Encoding	Channel	Eff
Position x	Horizontal axis	
Position y	Vertical axis	
Colour hue	Categorical groups	1 categ
Colour value	Continuous intensity	1 continuous (
Size	Bubble diameter	1 continuous (3–5 dist
Shape	Point marker	1 catego
Facet	Small multiples	
Total per chart		**5–**

Beyond 5 variables on a single chart, add **faceting** or switch to a fundamentally different chart type (parallel coordinates, heatmap, PCA).

Multi-Encoded Scatter

Five Variables on One Scatter Plot



Building the Multi-Encoded Scatter

R

```r
ggplot(ecar, aes(
    x = FICO,              #
         position x
    y = Rate,              #
         position y
    color = Tier,          # colour
    size = Amount,         # size
    shape = CarType)) + # shape
  geom_point(alpha = 0.4) +
  scale_color_viridis_c() +
  scale_size(range = c(1, 8)) +
  scale_shape_manual(values =
    c(N = 16, U = 17)) +
  theme_minimal() +
  labs(title = "5 Variables,
    1 Scatter")

# Marginal distributions
library(ggExtra)
ggMarginal(p, type = "histogram")
```

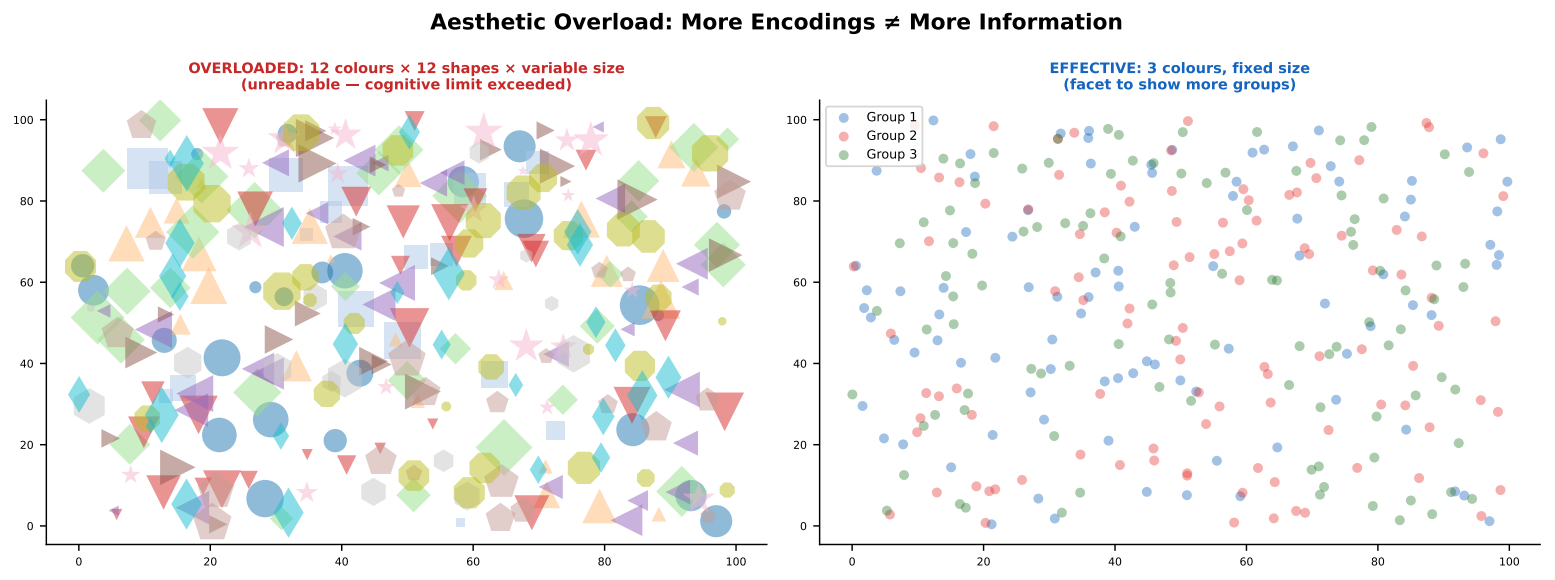
Python

```python
fig, ax = plt.subplots()
for ct, marker in [("N","o"),("U"
     ,"s")]:
    mask = ecar["CarType"] == ct
    sc = ax.scatter(
        ecar.loc[mask, "FICO"],
        ecar.loc[mask, "Rate"],
        c=ecar.loc[mask, "Tier"],
        s=ecar.loc[mask, "Amount"
            ]/300,
        marker=marker,
        alpha=0.4, cmap="viridis"
            ,
        label=ct)
plt.colorbar(sc, label="Tier")
ax.legend(title="Car Type")

# Marginal: seaborn jointplot
sns.jointplot(data=ecar,
    x="FICO", y="Rate",
    hue="CarType",
    kind="scatter", height=6)
```

Aesthetic Overload

More Encodings ≠ More Information

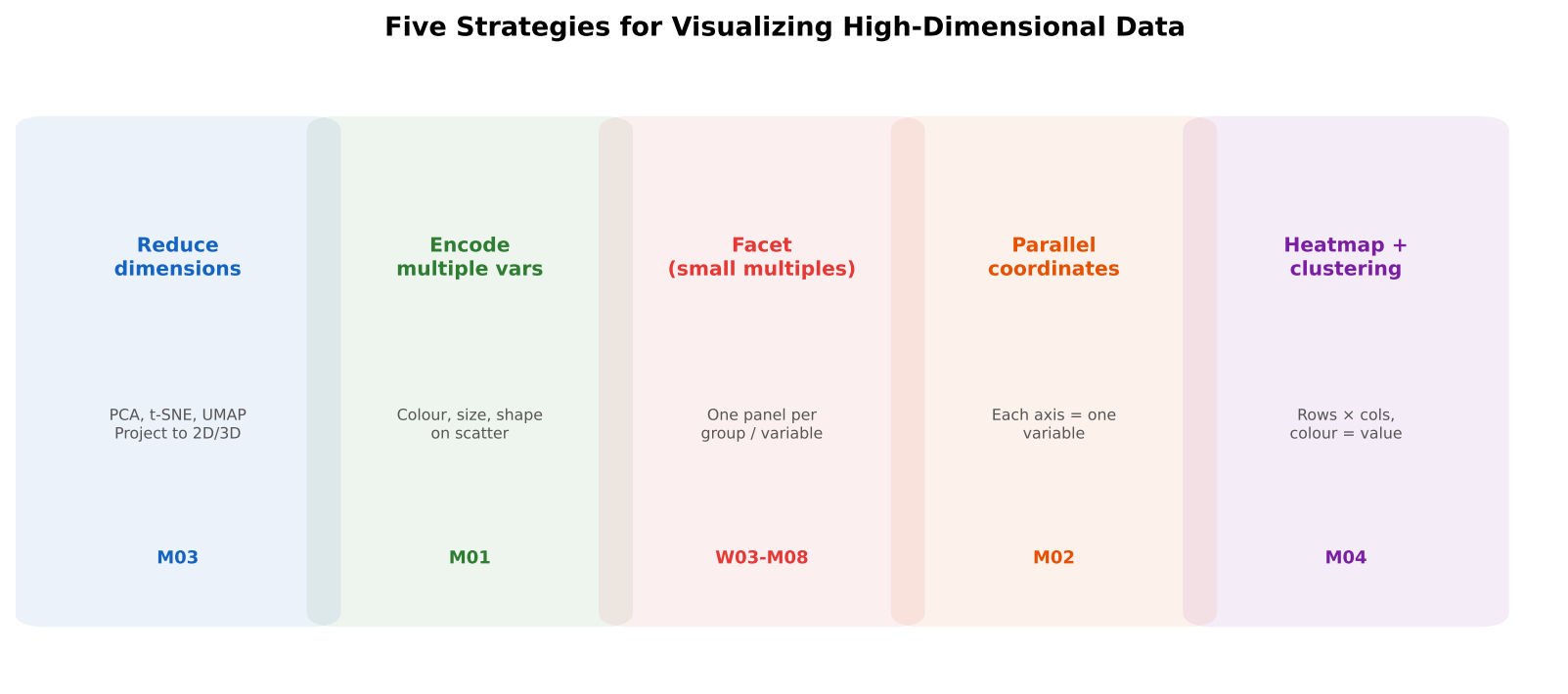


Pro-Tip

The human visual system can track ∼4 encoding channels simultaneously (Healey & Enns, 2012). Beyond that, the chart becomes a puzzle rather than a communication tool. **Cognitive limit**: ≤8 colour hues, ≤6 shapes, ≤5 size levels. If you need more groups, **facet** instead of encoding.

Five Strategies

Strategies for High-Dimensional Visualization



Choosing a Strategy

Situation	Strategy	Module
$\leq$5 variables	Multi-encoded scatter	This module
5–10 variables	Parallel coordinates	M02
5–10 variables (ranks)	Radar / spider chart	M02
10–100 variables	PCA / t-SNE / UMAP	M03
Variables $\times$ observations matrix	Heatmap + clustering	M04
Geographic data	Maps (choropleth, point)	M05–M08
$>$5 categorical groups	Small multiples (facet)	W03-M08

Aggregation for Large n

When n Is Too Large for Scatter

Five Aggregation Strategies for Large or High-Dimensional Data

Strategy	Tool	Mechanism	Effect
Binning	`Histogram, hexbin`	Continuous → discrete bins	Reduces n, reveals shape
Grouping	`group_by + summarise`	Per-category statistics	Collapses rows to groups
Sampling	`slice_sample(n=2000)`	Random subset for plotting	Scatter with 200K rows → 2K
Smoothing	`geom_smooth, LOESS`	Local regression through cloud	Reveals trend in noise
Marginals	`ggMarginal, jointplot`	Histograms on scatter margins	2D + 1D distributions

Aggregation in Practice

R

```r
# Hexbin (for 50K+ points)
ggplot(ecar, aes(x = FICO,
    y = Rate)) +
  geom_hex(bins = 50) +
  scale_fill_viridis_c() +
  theme_minimal()

# Sample + scatter
ecar |> slice_sample(n = 2000) |>
  ggplot(aes(x = FICO, y = Rate))
       +
  geom_point(alpha = 0.1, size
      =0.5)

# Smooth (LOESS through cloud)
ggplot(ecar, aes(x=FICO, y=Rate))
     +
  geom_smooth() +
  theme_minimal()

# Marginal histograms
library(ggExtra)
p <- ggplot(ecar, aes(x=FICO,
    y=Rate)) + geom_point(alpha
```

Python

```python
# Hexbin
ax.hexbin(ecar["FICO"], ecar["
     Rate"],
    gridsize=50, cmap="viridis",
    mincnt=1)
plt.colorbar(label="Count")

# Sample + scatter
sub = ecar.sample(2000)
ax.scatter(sub["FICO"], sub["Rate
     "],
    s=3, alpha=0.1)

# 2D KDE
sns.kdeplot(data=ecar, x="FICO",
    y="Rate", fill=True,
    cmap="Blues", levels=10)

# Jointplot (marginals included)
sns.jointplot(data=ecar,
    x="FICO", y="Rate",
    kind="hex", height=6)
```

Synthesis

Key Takeaways

1. The **curse of dimensionality**: same n in more dimensions = sparser coverage. Standard scatter fails beyond $\sim$3 variables.
2. **Encoding channels** are ranked: position > length > angle > area > colour hue > saturation > shape. Use the highest-ranked channels for the most important variables.
3. **Budget**: a single scatter can encode $\sim$5 variables (x, y, colour, size, shape). Beyond that, **facet** or switch chart types.
4. **Aesthetic overload** (>8 colours, >6 shapes) overwhelms the visual system. Simpler > denser.
5. Five strategies: **reduce** (PCA), **encode** (multi-scatter), **facet** (small multiples), **parallel** coordinates, **heatmap** + clustering.
6. For large n: **hexbin**, **sample**, **smooth**, **2D KDE**, **marginal** distributions.

Essential Readings

- Cleveland, W. S. & McGill, R. (1984). "Graphical Perception." *JASA*, 79(387). (Encoding channel ranking.)
- Healey, C. G. & Enns, J. T. (2012). "Attention and Visual Memory in Visualization." *IEEE TVCG*, 18(7).
- Munzner, T. (2014). *Visualization Analysis and Design*. CRC Press. (Channel effectiveness, task taxonomy.)
- Wilke, C. O. (2019). *Fundamentals of Data Visualization*, Chapter 2 (Encoding Channels).

Next Module: Parallel Coordinates & Radar Charts (W05-M02)